INCH-POUND
MIL-DTL-24643/14E
22 August 2002
SUPERSEDING
MIL-C-24643/14D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSSSGU
(INCLUDING VARIATION LSSSGA)

This specification is approved for the use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualifications required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter).
Third - Sealed binder (at manufacturers option).

Construction variant for type LSSSGU

Fourth - Cross-linked polyolefin jacket. (see table I for thickness.)

Construction variant for type LSSSGA

Fourth - Cross-linked polyolefin jacket. (see table I for thickness).
Fifth - Braided metal armor.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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TABLE I. Details.

Military Part no. M24643/14 1/	Size 2/	Conductor size		Type's Overall diameter (inches)				Dia over insulation nominal (inch)	Conductor resistance per 1000 ft at 25 °C max (ohms)	Jacket thickness min avg (inch)	Insulation resistance (megohms)	Cold bending mandrel dia (max) (inches)	Accelerated service loading amperes
				LSSSGU		LSSSGA							
		Navy standard	AWG	min	max	min	max						
-01UN	50		3 (class C)		0.520		0.570	0.334	0.209	0.040	160	4	240
-02UN	75		1 (class C)		0.602		.652	0.407	0.132	0.040	135	8	315
-03UN	100		0 (class D)		0.669		.719	0.453	0.105	0.050	125	9	375
-04UN	200		0000 (class D)		0.872		.922	0.634	0.0525	0.050	90	11	580
-05UN	300		300 MCM (class D)		1.001		1.051	0.748	0.0370	0.050	80	13	750
-06UN	400	400(127)			1.118		1.168	0.862	0.0268	0.050	70	14	925
-07UN	650		650 MCM (class D)		1.371		1.421	1.056	0.0170	0.060	60	17	1300
-08UN	800		800 MCM (class D)		1.485		1.535	1.170	0.0138	0.060	50	18	1550
-09UN	1000		1000 MCM (class D)		1.620		1.670	1.300	0.0110	0.060	40	20	---
-10UN	1600		1600 MCM (class B)		2.010		2.060	1.625	.00693	0.075	40	25	--
-11UN	2000		2000 MCM (class B)		2.210		2.260	1.810	0.00555	0.075	35	27	--

1/ Unarmored. When armored option is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LSSSGU and LSSSGA.

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EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - conductor to ground, volts, root mean square, minimum	5000
Insulation resistance - conductor to ground, Megohms -1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LSSSGA only) - conformance to material construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C, and D, plus the following:

Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and Size	Feet (nominal)
LSSSGU - 50 through 100	1500
LSSSGU - 200 through 1000	1000
LSSSGU - 1600 through 2000	500

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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Preparing Activity:
Navy - SH
(Project 6145-2308-012)

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